

Lab 16: Pseudocode Practice (Pair Work) - 5 pts

Work with your partner to write pseudocode for each problem below. Focus on your **logic and steps** — not Java syntax.

Do as many problems as you can. Get each problem checked by Mr.M before moving on to the next one. You will be graded on effort, not how many problems you complete.

1. Grade Classifier

A student's score (0–100) is given. Print their letter grade:

- 90 and above → A
 - 80–89 → B
 - 70–79 → C
 - 60–69 → D
 - Below 60 → F
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2. Find the Largest

Given an array of numbers, find and print the largest value.

3. Shopping Cart Total

You have an ArrayList of item prices (doubles). Calculate the subtotal, then apply a 10% tax to get the final total. Print both.

4. Name Formatter

Ask the user to enter their full name (first and last, separated by a space). Split it, then print:

```
1 | Last, First
```

For example, "Ada Lovelace" → "Lovelace, Ada"

5. Odd or Even Counter

Given an array of integers, count how many are odd and how many are even. Print both counts.

6. Countdown Timer

Ask the user for an integer `n`. Then, print every number from `n` down to 1, sleeping 1 second between each number. At the end, print "Blastoff!"

7. Password Strength Checker

Prompt the user to enter a password. Inform them of the requirements listed below. Then, check all of the requirements. If the password is not valid, tell the user that the password is invalid, and prompt them to enter another one.

- At least 8 characters long
- Contains the `"#"` character